

Value Stack Funnel

How the editor turns a battery on a site map into a value stack, gate by gate, and the five points where DividendVPP data would make that stack accurate in each market.

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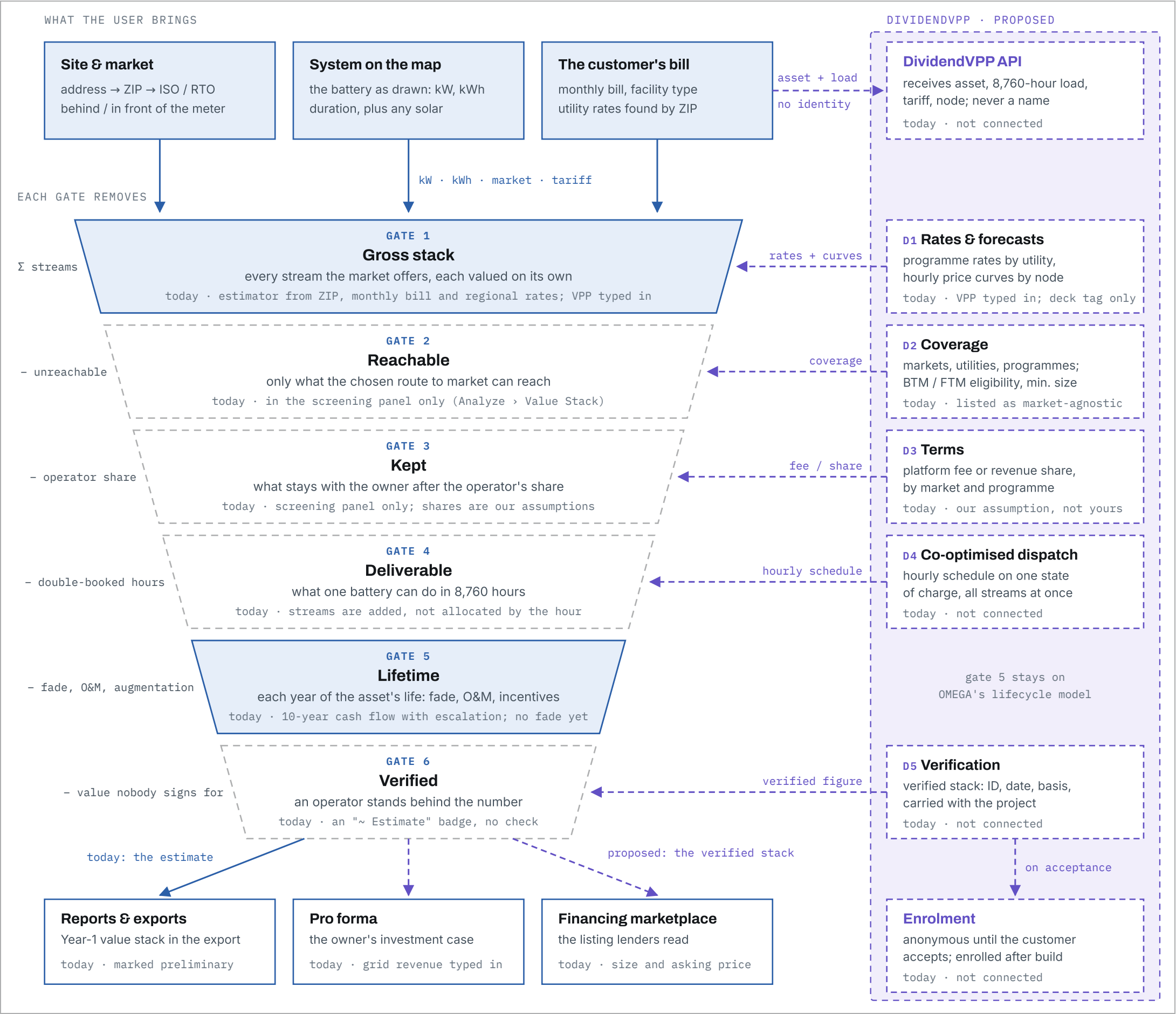
Every revenue stream a market offers goes in at the top. Each gate then takes out value that **this battery, in this market, through this route to market** cannot actually earn, so what leaves the bottom is a number someone can sign for.

Today the number a customer sees passes through two of the six gates: a first-pass estimate and a ten-year cash flow. The other four exist in places but are not yet applied to it. DividendVPP can supply market data, terms, an hourly dispatch and verification at five points, marked **D1 to D5**.

- applied to the number a customer sees today
- not yet applied to that number
- DividendVPP would supply it (proposed)

today · how the editor does it now

Figure 1. The six gates a value stack should pass through, and the five points where DividendVPP would feed them. Every violet exchange would run server to server, between OMEGA's /api and DividendVPP's API, so no DividendVPP credential reaches a browser and no customer identity leaves OMEGA before the customer accepts.



How a user moves through it

The same funnel as the screens a developer, installer or OEM opens in the editor, in the order they open them.

STEP 1

Put the battery on the map

Build › BESS Build › BESS Sizer › Solar + Storage

The value stack reads kW and kWh off the placed fleet, then the configured list. No battery drawn, no stack.

feeds: system on the map

STEP 2 › OPTIONAL

Screen the market

Analyze › Screening › Value Stack

ZIP to market, behind or in front of the meter, and who monetises it: self, a utility programme, an aggregator or Molecule Systems. A screening view; its figures stay in the panel.

gates 1–3, screening only

STEP 3

Estimate the streams

Estimate › Cost & ROI › Construction Cost › Revenue

ZIP (prefilled), the customer's monthly bill and facility type, plus a regulation price in PJM, then Estimate Value Stack. A front-of-meter project is turned away here.

gate 1

STEP 4

Apply and fill the gaps

Revenue › Apply to Revenue Streams

Arbitrage, peak demand, demand response and PJM regulation fill in with an "~ Estimate" badge. Virtual power plant, capacity and resilience are typed in by hand.

gate 1

STEP 5

Read the return

Cost & ROI › ROI › results rail

Payback, year-one return, and a ten-year NPV and IRR, with revenue escalated at a fixed rate.

gate 5

STEP 6

Present and hand off

Export Analysis › Output › Marketplace

The analysis export prints an Estimated Year-1 Value Stack marked preliminary. Push to Marketplace carries the size and asking price, and Apply for Financing the project itself; neither carries the stack.

outlets

Market by market

What a battery can be paid for in each market, and how the number that reaches the customer is valued today. Each amber, blue or outlined cell is a place DividendVPP data would change the answer.

Estimate

 regional planning figure

Typed in

 no rate on file; the user enters one

Not valued

 the market pays for it; the model doesn't count it

Coming

 built, not open yet

No market

 nothing to earn here

MARKET	CAPACITY & DEMAND RESPONSE	ANCILLARY SERVICES	VPP PROGRAMME	ENERGY ARBITRAGE	WHAT DIVIDENDVPP DATA WOULD SHARPEN
PJM Mid-Atlantic, Ohio Valley, ComEd	Estimate one band per kW-year	Estimate regulation at an editable clearing price	Typed in	Estimate retail time-of-use spread	Rates for the utility VPP and demand-response programmes you enrol into; node energy and regulation forecasts
ERCOT Texas, energy-only	No market	Not valued Coming the ERCOT price deck takes all five services	Typed in	Estimate Coming retail spread today; nodal via the price deck	Hourly DA, RT, Reg-Up, Reg-Down, RRS, ECRS and Non-Spin at the settlement point: the seven series the price deck already takes
CAISO California	Estimate resource adequacy is bilateral	Not valued	Typed in	Estimate	Programme rates by utility; RA values; ancillary forecasts
NYISO New York	Estimate ICAP varies by zone	Not valued	Typed in	Estimate	ICAP by zone; VDER and programme rates; ancillary forecasts
ISO-NE New England	Estimate	Not valued	Typed in	Estimate	Programme rates by state and utility, with event history; ancillary forecasts
MISO Central US	Estimate	Not valued	Typed in aggregated DR opening state by state	Estimate	Which utilities allow aggregated demand response, and what it pays
SPP Southern Plains	Estimate a default band	Not valued	Typed in	Estimate	Programme availability; ancillary forecasts
Outside an ISO Southeast, West	Estimate a default band	No market	Typed in	Estimate utility time-of-use spread	Utility programmes you enrol assets into

Bill-side value, demand charges and time-of-use savings, comes from the customer's own monthly bill in every market.

The screening panel (Analyze › Value Stack) goes further in places: PJM capacity at the published auction price, ancillary services for front-of-meter assets, and named programmes such as ConnectedSolutions, SGIP and NYSERDA. Its figures stay in that panel today.

The market is resolved from the site's state. Where a state straddles two markets (Illinois, Indiana, Michigan, Kentucky, North Carolina), the utility territory or pricing node should decide, which is part of what D2 coverage settles.

What DividendVPP would send, and in what shape

Each point lands on one gate. The shapes follow rules the editor already applies to any price data it takes in.

POINT	WHAT	SHAPE OMEGA NEEDS	WHERE IT LANDS
D1 Rates & forecasts	Rates for the programmes you enrol assets in; hourly price forecasts	Rate, unit (per kW-year, per kW-summer, per event), programme, utility, source and effective date. Forecasts as 8,760 hourly rows per product per settlement point, unit declared (\$/MWh or \$/MW-h), negative prices kept, a vintage label.	Gate 1. Fills the VPP line, typed in today, and the programme and capacity rates, each with its source. A forecast becomes a price deck locked to its vintage.
D2 Coverage	Markets, utilities and programmes you can enrol into	Per programme: market, utility territory, behind or in front of the meter, minimum kW, telemetry, season.	Gate 2. Which streams the DividendVPP route reaches, resolved by utility territory rather than by state.
D3 Terms	Platform fee or revenue share	Per market and programme, with any minimums.	Gate 3. What stays with the owner on the DividendVPP route.
D4 Dispatch	A co-optimised hourly schedule	Request: kW, kWh, round-trip efficiency, warranty cycles, 8,760-hour load, tariff, node, and no customer name or address. Response: hourly charge, discharge and state of charge; revenue per stream; value forgone.	Gate 4. An hourly run on one state of charge instead of streams added together: the number a lender funds.
D5 Verification	A figure you stand behind	Verification ID, date, the assumptions behind it, and how long it holds.	Gate 6. Replaces the "~ Estimate" badge, and travels with the project to reports, the pro forma and the financing listing.

Enrolment would follow the partner route OMEGA already runs for equipment vendors. The vendor sees an anonymous opportunity (state, utility, size, stage), and the customer's identity is released only when the customer accepts.

Rules the editor already holds itself to

Anything DividendVPP sends would be held to the same rules.

The stack sizes off the drawing. kW and kWh come from the battery on the site map or its configured list, never from a number typed into the revenue form. No battery, no stack.
A missing rate is shown, not hidden. A stream the market doesn't have is marked n/a, one with no rate on file stays at zero until someone enters it, and a utility the rate table doesn't know is flagged "default, verify".
Estimates say so. Every applied stream carries an "~ Estimate" badge, every figure in the screening panel shows its arithmetic, and exports call the stack preliminary until it is verified.
Price imports ask for units. \$/MWh and \$/kWh differ by a thousand, so the price-deck importer asks instead of guessing, keeps negative prices (the hours a battery earns by charging) and locks a deck to its vintage so a case already sent cannot re-baseline.
A merchant asset is not given a retail answer. The retail estimator turns away a front-of-meter project instead of valuing it with retail rates it will never see.

Where things stand

IN USE TODAY - Estimate › Cost & ROI › Revenue: the estimator, applied to revenue streams, the ROI and the results rail - Analyze › Value Stack: a screening view by market and route to market, with Molecule Systems as one route - An analysis export with an Estimated Year-1 Value Stack, marked preliminary	BUILT, NOT OPEN YET - Price Decks: nodal energy and ancillary series by settlement point, with a "DividendVPP forecast" provider tag - An allocation engine that spends 8,760 hours and the warranty's cycles across streams, and a lifecycle model with fade and augmentation	PROPOSED HERE - D1 to D5, server to server between OMEGA's /api and DividendVPP's API - The verified stack carried to reports, the pro forma and the financing marketplace - To start, from DividendVPP: a sample forecast export, the coverage list, terms by market, and dispatch API documentation
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